

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Year: 2018

Subject: 392132 Physics II

Section: 15 - 18

Date: 28 Feb 2019

Time: 10:00-12:00

Name: _____ ID: _____ Field of Study: _____

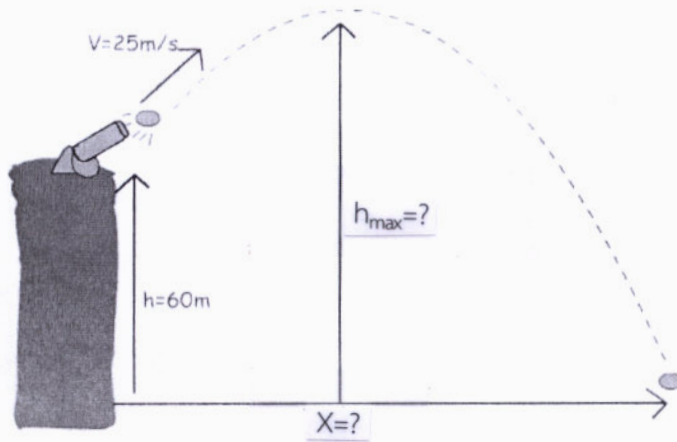
Instructions:

1. The examination has 7 pages (including this page) and a total score of 60 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are **NOT** allowed to leave the examination room during the first 1 hour after the beginning of the exam.
5. You are **NOT** allowed to open the exam papers or start to answer before the proctor's permission.
6. You are **NOT** allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Student who are caught cheating will also be denied registering for the following semester.
11. Express your answer in **English**, other languages will be discarded.
12. If not specified otherwise, used these values: $g = 10 \text{ m/s}^2$, $\pi = 3.14$.

Score Report	
1	
2	
3	
4	
5	
6	
Total	

Cheating in the exam is considered an extremely serious Offence which will result in expulsion from the University

(1) In the given picture you see the motion path of a cannonball. Find the maximum height it can reach, horizontal distance it covers and total time from the given information. (10 points)



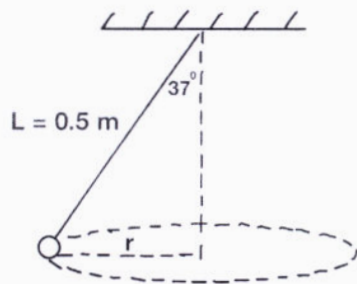
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Name _____ ID _____ Seat No. _____ 392132/3

(2) A rocket is launched at an angle of 53° above the horizontal with an initial speed of 100 m/s. The rocket moves for 3 s along its initial line of motion with an acceleration of 30 m/s^2 . At this time, its engines fail and the rocket proceeds to move as a projectile. Find the maximum altitude reached by the rocket (4 points), the total time of flight (3 points) and its horizontal range (3 points).

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(3.1) An object hanged on a rope $L = 0.5 \text{ m}$, does rotational motion. If the angle between rope and vertical is 37° . Find the magnitude of tangential velocity of the object. (4 points)

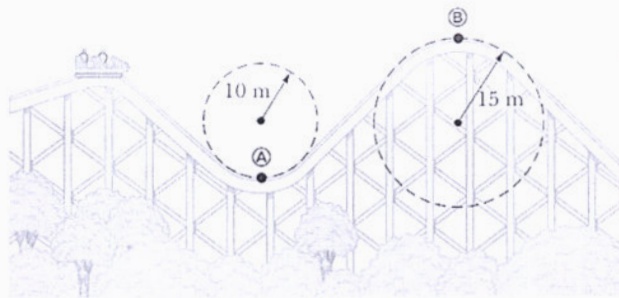


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(3.2) A race car accelerates uniformly from a speed of 36 m/s to a speed of 72 m/s in 3 s while traveling counterclockwise around a circular track of radius 500 m . When the car reaches a speed of 50 m/s . Find the magnitude of the car's centripetal acceleration (2 points), the angular speed (2 points) and the magnitude of the total acceleration (2 points).

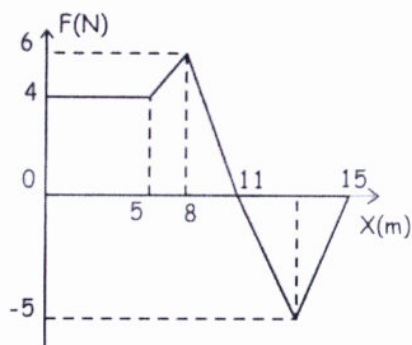
Name _____ ID _____ Seat No. _____ 392132/5

- (4) A rollercoaster vehicle has a mass of 500 kg when fully loaded with passengers.
- (a) If the vehicle has a speed of 20.0 m/s at point A, what is the force of the track on the vehicle at this point? (5 points)
- (b) What is the maximum speed the vehicle can have at point B for gravity to hold it on the track? (5 points)



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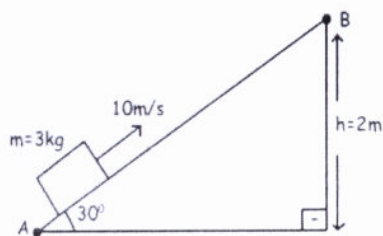
(5.1) Applied force vs. position graph of an object is given below. Find the work done by the forces on the object. (3 points)



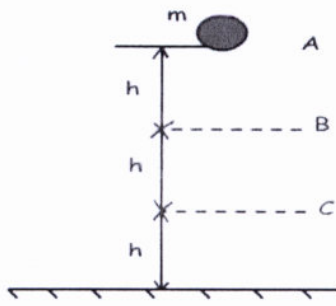
(5.2) A 200 g baseball is thrown so that it acquires a speed of 25 m/s.

- (a) What is its kinetic energy? (2 points)
- (b) What was the net work done on the ball to make it reach this speed, if it started from rest? (2 points)

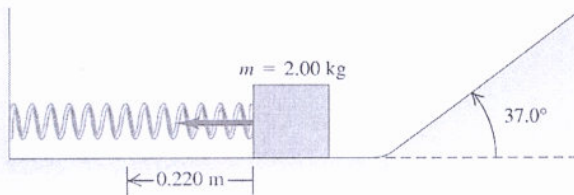
(5.3) Box having mass 3 kg thrown with an initial velocity 10 m/s on an inclined plane. If the box passes from the point B with 4 m/s velocity, find the work done by friction force. (3 points)



(6.1) An object does free fall. Picture given below shows this motion. Find the ratio of kinetic energy at point C to total mechanical energy of the object. (5 points)



(6.2) A 2.00-kg block is pushed against a spring with negligible mass and force constant $k = 200 \text{ N/m}$, compressing it 0.220 m. When the block is released, it moves along a frictionless, horizontal surface and then up a friction incline with a coefficient of kinetic friction is 0.2 and slope 37.0° as shown. What is the speed of the block as it slides along the horizontal surface after having left the spring? And how far does the block travel up the incline. (5 points)



Good Luck!!

Asst.Prof.Dr.Prakasit CHANGSUPAN

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